

GUNSAN, Korea, Republic of

WMO: 471400

Lat: 36.0053N

Lon: 126.7614E

Elev: 81

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	20.7	5.4	7.3	24.0	8.5	8.5	24.5	30.6	30.2	25.5	33.2	4.5	90	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	89.5	79.4	87.3	78.5	85.2	77.4	81.0	86.5	79.8	85.1	78.7	83.4	8.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	154.4	84.0	78.3	147.3	82.8	77.3	142.2	82.0	44.9	86.7	43.5	85.6	42.3	83.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.5	19.8	16.7	DB	13.0	93.0	4.2	2.1	10.0	94.5	7.5	95.7	5.1	96.8	2.0	98.3
			WB	11.5	81.9	3.9	1.7	8.7	83.2	6.4	84.2	4.2	85.1	1.4	86.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	31.9	34.9	42.7	53.2	63.2	71.4	78.0	79.3	71.2	60.4	48.2	36.3
	DBStd	17.27	6.24	6.05	5.99	6.17	4.66	3.40	3.97	4.12	4.42	5.51	6.72	6.70
	HDD50	1803	562	424	238	36	0	0	0	0	0	4	112	427
	HDD65	4569	1027	842	691	358	91	2	0	0	4	159	504	891
	CDD50	3988	0	1	12	130	408	641	867	908	636	326	58	1
	CDD65	1282	0	0	0	3	34	193	402	443	190	17	0	0
	CDH74	8921	0	0	0	15	128	690	3144	4057	858	29	0	0
	CDH80	2311	0	0	0	2	8	70	831	1283	116	1	0	0
Wind	WSAvg	6.3	6.0	6.7	7.8	7.5	6.9	6.1	6.3	5.9	5.5	5.5	6.0	5.7
Precipitation	PrecAvg	46.8	1.3	1.5	2.0	3.4	3.5	5.3	9.9	9.0	5.4	1.9	2.3	1.4
	PrecMax	69.6	3.6	3.9	7.4	8.4	7.4	12.8	23.8	24.1	11.4	4.7	6.2	2.6
	PrecMin	28.0	0.3	0.2	0.1	0.5	0.3	0.7	2.6	3.8	0.2	0.0	0.7	0.2
	PrecStd	10.8	1.0	1.1	1.4	1.9	1.8	3.2	5.3	4.3	3.6	1.3	1.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	59.7	65.9	75.8	80.8	84.8	91.4	93.2	86.1	77.0	68.8	56.4
		MCWB	45.8	53.0	55.6	62.4	64.7	70.5	79.2	80.2	75.0	66.4	60.3	49.8
	2%	DB	46.4	51.4	60.1	70.2	77.4	81.9	88.8	90.3	83.1	74.0	64.3	51.9
		MCWB	41.9	45.4	51.3	58.5	64.3	70.2	79.1	79.5	73.8	65.0	57.7	46.7
	5%	DB	43.4	47.1	56.3	66.7	74.8	79.9	86.7	88.6	80.9	71.5	61.4	48.7
		MCWB	39.2	42.0	49.2	56.6	63.2	70.2	78.5	79.2	72.1	63.3	55.3	44.1
	10%	DB	40.7	44.0	52.7	63.5	72.2	78.0	84.6	86.5	78.9	69.1	58.4	46.0
		MCWB	36.7	39.5	46.4	54.8	62.4	69.8	77.8	78.4	71.2	61.8	52.9	41.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.0	55.6	58.3	65.4	70.8	76.4	82.6	83.4	78.3	71.1	62.9	51.3
		MCDB	49.8	58.6	62.8	71.1	74.8	80.2	87.0	88.1	83.3	74.7	66.8	54.8
	2%	WB	42.8	46.4	54.2	61.6	67.7	74.4	80.7	81.7	76.2	67.6	59.3	47.8
		MCDB	46.0	49.8	58.7	67.4	73.5	78.2	85.9	87.1	80.7	71.0	62.8	51.0
	5%	WB	39.8	42.7	50.0	58.7	65.6	72.8	79.5	80.4	74.6	65.2	56.5	44.5
		MCDB	42.7	46.2	54.7	64.3	71.4	76.8	84.6	86.2	78.5	69.4	60.4	48.0
	10%	WB	37.2	40.2	47.0	56.2	64.0	71.5	78.4	79.3	72.9	63.2	53.7	42.1
		MCDB	40.3	43.5	51.6	61.5	69.6	75.6	83.2	84.3	77.2	67.3	57.8	45.8
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.6	14.1	15.3	14.7	12.0	9.7	10.8	12.6	14.6	13.5	12.1
		MCDBR	14.5	17.0	20.1	20.5	19.0	15.0	12.5	13.3	14.3	16.4	16.3	15.4
		MCWBR	11.0	12.6	12.9	10.6	8.8	6.5	5.3	5.3	6.8	8.6	10.9	11.5
	5% WB	MCDBR	13.3	15.6	18.2	17.0	15.4	11.1	11.1	12.1	11.8	13.5	14.1	14.5
		MCWBR	11.1	12.9	13.4	10.9	9.2	6.3	5.7	5.6	7.0	8.9	11.3	11.9
Clear-Sky Solar Irradiance	taub	0.454	0.493	0.570	0.593	0.611	0.681	0.630	0.574	0.503	0.486	0.472	0.453	
	taud	1.912	1.851	1.707	1.705	1.722	1.647	1.805	1.942	2.061	1.968	1.947	1.927	
	Ebn at Noon	219	228	222	227	226	210	220	230	239	228	212	209	
	Edh at Noon	47	56	71	75	74	80	68	58	49	49	45	44	
All-Sky Solar Radiation	RadAvg	757	1020	1384	1605	1772	1616	1399	1444	1321	1159	795	663	
	RadStd	54	90	132	127	178	151	196	210	134	93	89	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+1.10	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	+76

Nomenclature: See separate page